

Convergence to Equilibrium of Stochastic Exchange Models

Matteo Quattropani



Discrete Random Structure II — MAY 18, 2026

Based on joint works with **Pietro Caputo** and **Federico Sau**

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Initiated by Marc Kac in the '50s

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- **Mixing time:**

$$\forall \varepsilon \in (0, 1), \quad t_{\text{mix}}(\varepsilon) = \inf\{t \geq 0 \mid d_{\text{TV}}(t) \leq \varepsilon\}$$

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- Take a sequence of Markov chains on a state space $|\Omega_n| \rightarrow \infty$

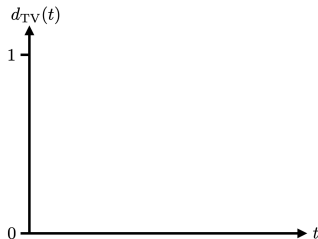
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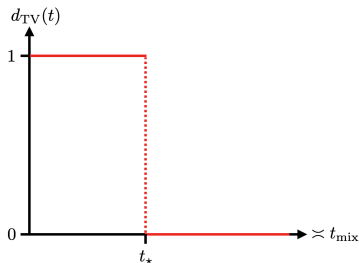


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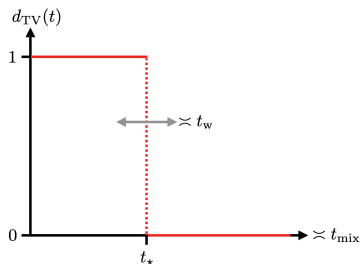


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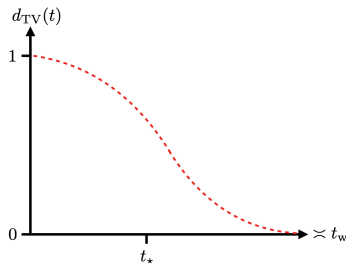


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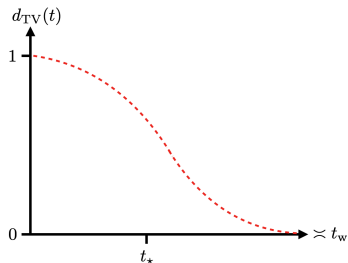


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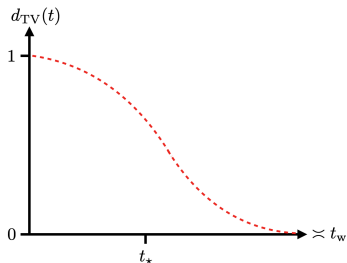
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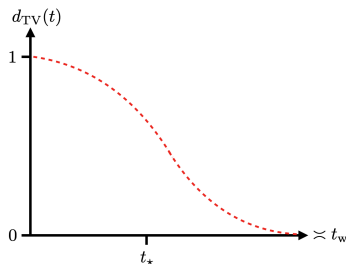
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- Continuous state-space Markov chains: much less developed

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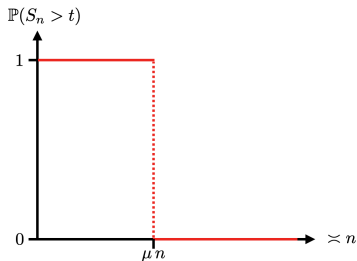
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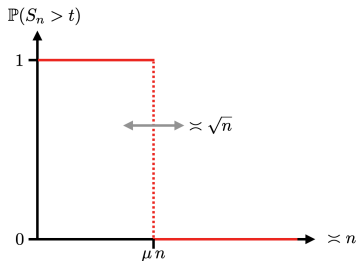
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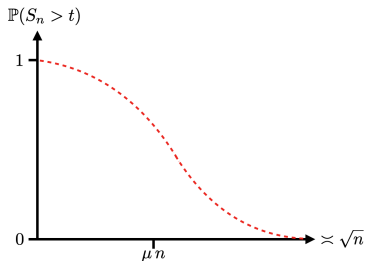
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Q: How fast does the **random** function $t \mapsto d_{\text{TV}}^{\text{avg}}(\eta_0, t)$ decay?

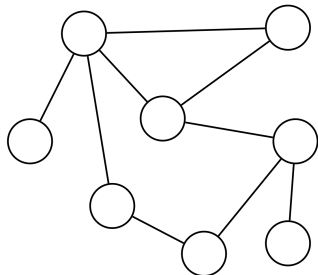
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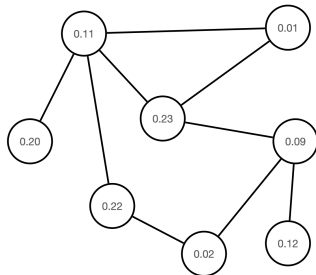
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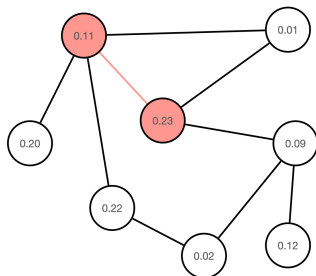
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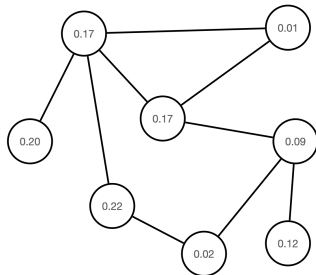


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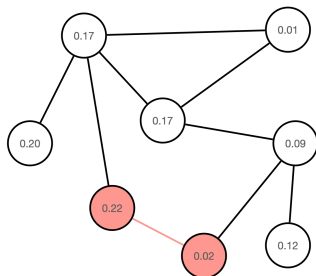


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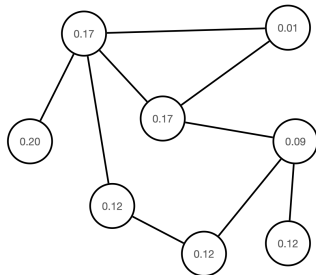


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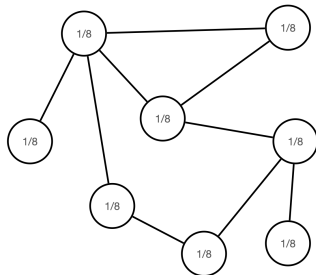


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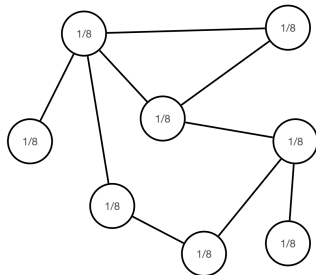
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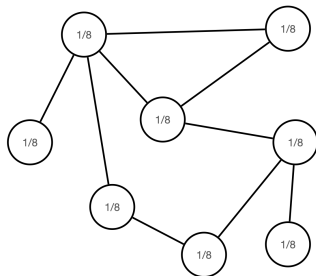
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Remark

- For any $\eta_0 \in \Delta$ the function $t \mapsto d_{TV}^{avg}(\eta_0, t)$ is a.s. non-increasing.

AVG on graphs

Natural extension: introduce an underlying **geometry** of interactions



- A graph $G([n], E)$
- **Initial energy configuration:**
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Remark

- For any $\eta_0 \in \Delta$ the function $t \mapsto d_{\text{TV}}^{\text{avg}}(\eta_0, t)$ is a.s. non-increasing.
- We are mostly interested in *worst-case* analysis

$$d_{\text{TV}}^{\text{avg}}(t) = \max_{\eta_0 \in \Delta} d_{\text{TV}}^{\text{avg}}(\eta_0, t).$$

- It can be shown that the maximum can be taken over Dirac distributions on $[n]$.

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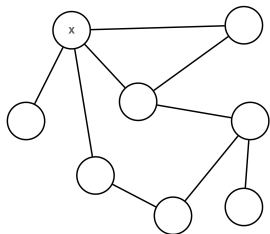
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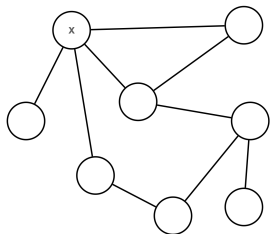
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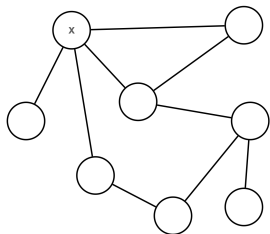
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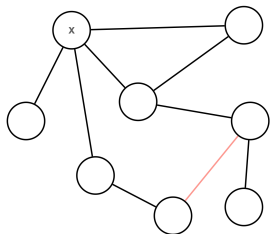
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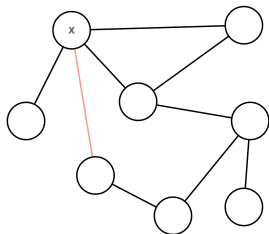
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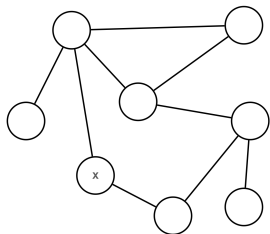
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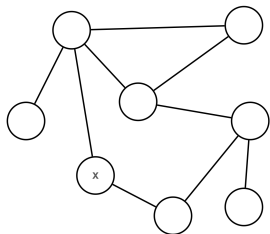
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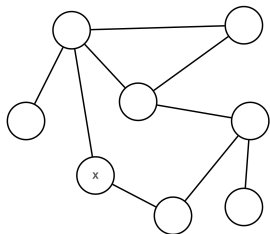
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$(\eta_t)_{t \geq 0}$ is a *noisy version* of the law of the RW $(\eta_0 P_t)_{t \geq 0}$

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M.Q., **F. Sau**, '21

P. Caputo, **M.Q.**, **F. Sau**, '22, '24, '25, '26

R. Movassagh, **M. Szegedy**, **G. Wang**, '22

F. Sau, '23, '24

A. Eide, '24

N. Gantert, **T. Vilkas**, '24

D. Elboim, **Y. Peres**, **R. Peretz**, '25

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A. Campos, **T. Franco**, **M. Heydenreich**, **M. Schrocke**, '25

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Answer to Bourgain's question

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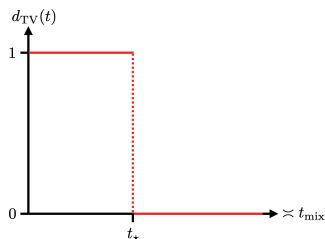
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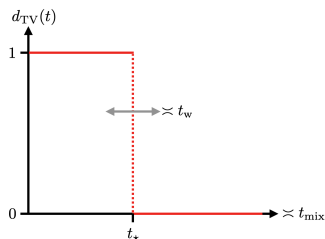
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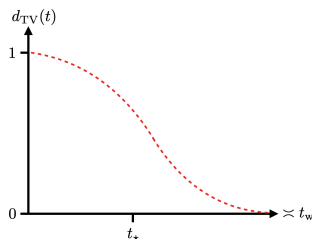
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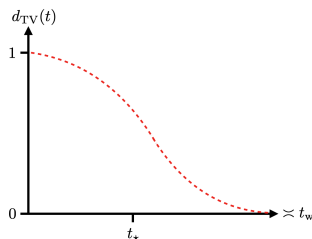
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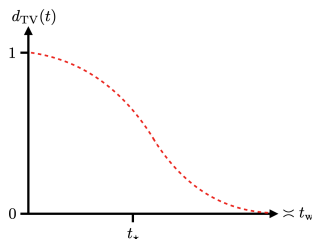
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- **RW** does not exhibit cutoff, but **AVG** does
- The convergence occurs on **different scales**:

$$t_{\text{mix}}^{\text{RW}} \asymp \frac{1}{n} \quad \text{while} \quad t_{\text{mix}}^{\text{AVG}} \asymp \frac{\log n}{n}$$

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Graph	RW	AVG	scale	Ref.
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Open Question: is there a graph on which...

- **RW** exhibits cutoff but **AVG** does not?
- neither **RW** nor **AVG** exhibit cutoff, but at **different scale**?
- both **RW** and **AVG** exhibit cutoff, but at a **different scale**?

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Let $d \geq 3$ fixed and G be a typical realization of a d -regular random graph.

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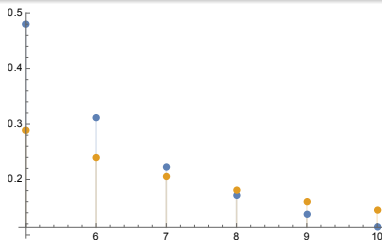
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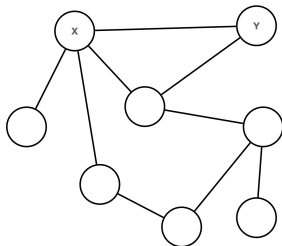


If **AVG** has cutoff on **scale** $\log n$, the cutoff point must be **larger** than the one of the **RW** if $d \geq 8$!

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L^2 upper-bound

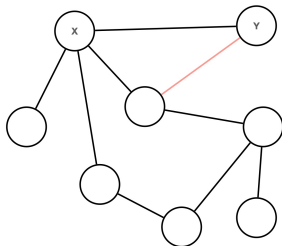
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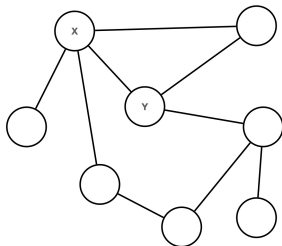
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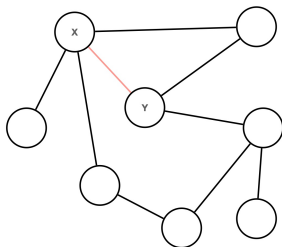
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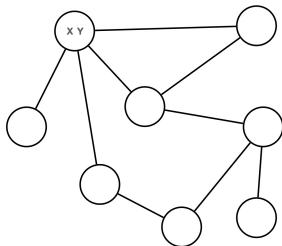
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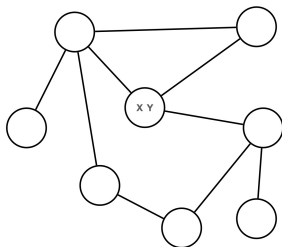
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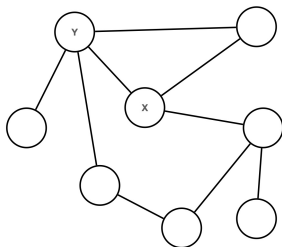
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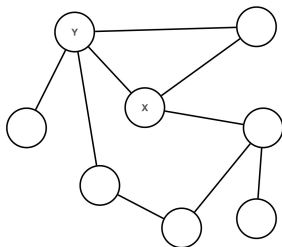
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$$2\mathbb{E}[\|\eta_t - \pi\|_{\text{TV}}] \leq \mathbb{E} \left[\left\| \frac{\eta_t}{\pi} - 1 \right\|_{L^2(\pi)}^2 \right] = n \mathbb{P}_{\eta_0 \otimes \eta_0}^{\text{CRW}}(X_t = Y_t) - 1$$



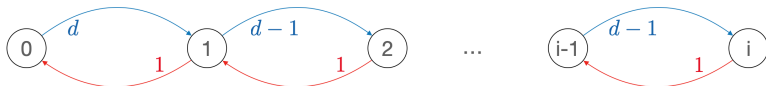
Standard idea:

Approximate $\mathbb{P}_{\delta_x \otimes \delta_x}^{\text{CRW}}(X_t = Y_t)$ with its analogue on a d -regular tree.

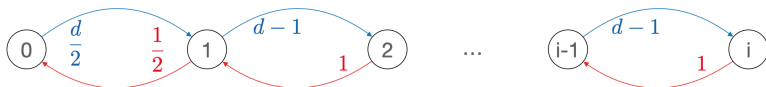
$\Rightarrow$ Study a B&D chain!

AVG on the Random Regular case: upper bound

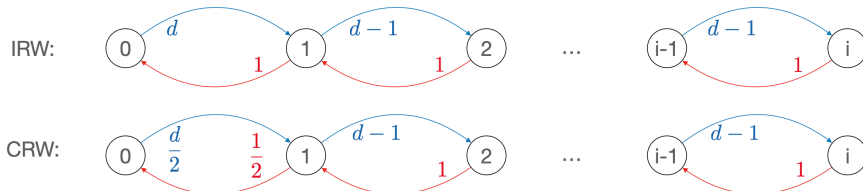
IRW:



CRW:



AVG on the Random Regular case: upper bound

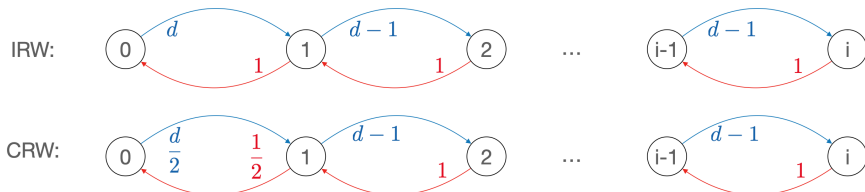


IRW vs CRW on an infinite d -ary tree

$$\mathbb{P}_{\delta_x \otimes \delta_x}^{\text{IRW}}(X_t = Y_t) \approx \frac{1}{n} \iff t = \frac{\log n}{d(1-\varrho)} \quad \varrho := \frac{2\sqrt{d-1}}{d}$$

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Consequences:

- Cutoff in the $L^2(\pi)$ distance at time $t_\star = \frac{\log n}{d(1-\gamma)}$
- The L^2 -cutoff time of RW and AVG coincide if and only if $d \leq 10$
- $\mathbb{E}[\|\eta_x - \pi\|_{\text{TV}}] \approx 0$ if $t \geq t_\star$. Hence, precutoff!

AVG on the hypercube

Consider the d -dimensional hypercube $[n] = \{0, 1\}^d$.

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Both in TV and in L^2 there is cutoff at

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- **Left to prove:** the “defected edge” has no first-order impact, i.e.,

$$\mathbb{P}_{\eta_0 \otimes \eta_0}^{\text{CRW}}(X_t = Y_t) \approx \mathbb{P}_{\eta_0 \otimes \eta_0}^{\text{IRW}}(X_t = Y_t), \quad t \approx t_\star$$

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Both **RW** and **AVG** exhibit cutoff at the very **same time!**

AVG on finite dimensional graphs

AVG on finite dimensional graphs

A sufficient condition for the absence of RW cutoff

Suppose that the RW satisfies

$$\exists C_1, C_2 > 0 \quad \text{s.t.} \quad \limsup_{n \rightarrow \infty} \max_{y \in [n]} \max_{x \in [n]} \frac{P_t(x, y)}{\pi(y)} \leq C_1, \quad t \geq C_2 t_{\text{rel}}$$

Then

$$“t_{\text{mix}} = \Theta(t_{\text{rel}})” + “\text{product condition}” \implies \text{No cutoff for RW}$$

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Nash inequality

There exist (d, t_{Nash}, T) such that, as $n \rightarrow \infty$,

$$d = O(1), \quad t_{\text{Nash}} = O(t_{\text{rel}}), \quad T = \Theta(t_{\text{Nash}}) \quad (\text{FD})$$

$$\forall f : \|f\|_1 = 1, \quad \|f\|_2^{2 + \frac{4}{d}} \leq t_{\text{Nash}} \left(\mathcal{E}_{\text{RW}}(f) + T^{-1} \|f\|_2^2 \right) \quad (\text{Nash})$$

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AVG on finite dimensional graphs

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Neither **RW** nor **AVG** cutoff, but they mix **on the same timescale!**

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- This talk: the **Kipnis-Marchioro-Presutti (KMP) model**

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Q: Which metric should we use to quantify the convergence?

Convergence in “true” Total-Variation

TV-cutoff on the segment (Caputo, Labbé, Lacoïn, '20)

For all $\alpha \geq 1$, the $\text{KMP}(\alpha)$ on the n -segment satisfies

$$d_{\text{TV}}^{\text{KMP}(\alpha)}(cn^2 \log n) \sim \begin{cases} 1 & \text{if } c < \frac{1}{\pi^2} \\ 0 & \text{if } c > \frac{1}{\pi^2} \end{cases}$$

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TV-precutoff in mean-field (Smith, '14)

For all $\alpha > 0$, there exists $C_-(\alpha) < C_+(\alpha)$ such that the $\text{KMP}(\alpha)$ on the complete graph on n -vertices satisfies

$$d_{\text{TV}}^{\text{KMP}(\alpha)}\left(c \frac{\log n}{n}\right) \sim \begin{cases} 1 & \text{if } c < C_- \\ 0 & \text{if } c > C_+ \end{cases}$$

Wasserstein distances

For the AVG process we considered

$$\mathbb{E} \left[\|\eta_t - \pi\|_{\text{TV}} \right] = \frac{1}{2} \sum_{x \in [n]} \mathbb{E} \left[\left| \eta_t(x) - \frac{1}{n} \right| \right]$$

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The latter coincides with

$$W_1(\eta_0, t) = \inf_{\substack{\xi \sim \eta_t \\ \xi' \sim \eta_0}} \mathbb{E} [\|\xi - \xi'\|_{\text{TV}}]$$

in the case of the AVG with $\eta_0 \in \Delta$, in which case

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Weaker than TV

$$W_1(\eta_0, t) \leq d_{\text{TV}}(\text{Law}(\eta_t), \text{Law}(\eta_\infty)) = \frac{1}{2} \inf_{\substack{\xi \sim \eta_t \\ \xi' \sim \eta_\infty}} \mathbb{E} [\mathbf{1}\{\xi \neq \xi'\}]$$

KMP: cutoff in W_1 at the entropic time on the complete graph

Recall $U \sim \text{Beta}(\alpha, \alpha)$

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$$\widehat{U} \quad \text{size-biased of } U \quad \mathbb{P}(\widehat{U} \leq u) = 2 \mathbb{E}[U \mathbf{1}_{U \leq u}]$$

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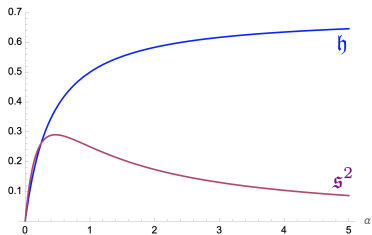
Cutoff and profile for the mean-field KMP (Caputo, Q., Sau '25)

For any $\beta \in \mathbb{R}$

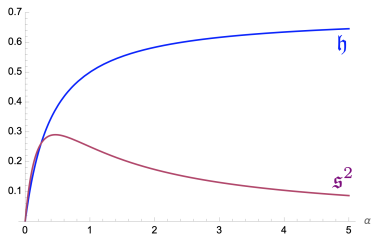
$$\lim_{n \rightarrow \infty} W_1(\mathbf{1}_{x_0}, t_{\text{ent}} + \beta t_{\text{w}}) = \Phi \left(-\frac{\beta(1 + \mathfrak{r})}{\sqrt{1 + \mathfrak{r}^2}} \right)$$

where Φ is the standard normal CDF.

Comments

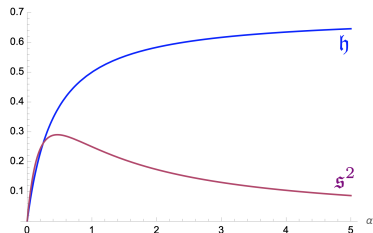


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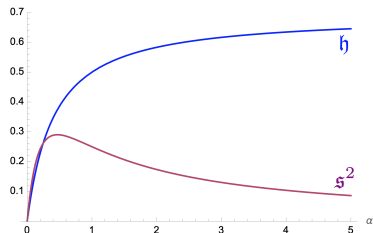
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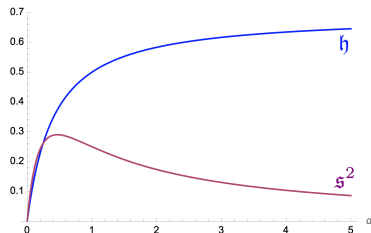
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- W_1 -mixing same timescale as TV-mixing [Smith '14]

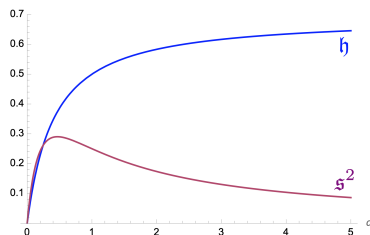
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Why “entropic”?

Naive: $h = -\mathbb{E}[\log(\widehat{U})] = -2\mathbb{E}[U \log U] \leq \log(2)$

Phenomenological: toward-equilibrium behavior is in line with the “entropic time picture” developed for finite Markov chains in the last decade

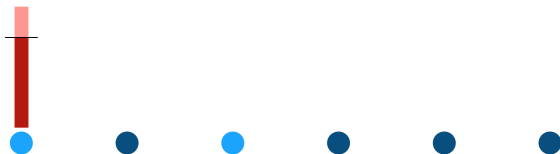
The pile dynamics



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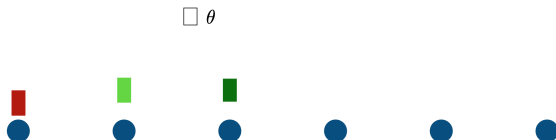
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Left to prove: $W_1(t) \approx \mathbb{E}[\|\eta_t^{\geq \theta}\|_1]$

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THANK YOU FOR THE ATTENTION!